

Ideation

Project Overview

EduGenie is an AI-powered educational assistant designed to enhance digital learning by integrating Google's Gemini AI with a FastAPI backend and an interactive web interface.

Motivation

Modern education requires intelligent systems that provide instant responses, personalized learning, and efficient content summarization.

Features

- AI Question Answering
- Concept Explanation
- Quiz Generation
- PDF Summarization
- Learning Path Recommendation
- Voice Input
- Text-to-Speech
- Dark/Light Mode
- Copy Response
- Download as PDF/TXT
- Responsive UI

Technology Stack

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	FastAPI
Language	Python
AI Model	Google Gemini API
PDF Processing	PyPDF2

Component	Technology
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Styling	CSS
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Innovation

Unlike traditional educational websites, EduGenie integrates multiple AI services into one platform, enabling students to ask questions, summarize PDFs, generate quizzes, and receive learning recommendations without switching between different applications.

Future Enhancements

- User authentication.
- Learning progress tracking.
- Multi-language support.
- AI-generated flashcards.
- Cloud deployment.
- Student performance analytics.